

THE ROLE OF RATE OF FORCE DEVELOPMENT ON VERTICAL JUMP PERFORMANCE

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ABSTRACT

McLellan, CP, Lovell, DI, and Gass, GC. The role of rate of force development on vertical jump performance. *J Strength Cond Res* 25(2): 379–385, 2011—The purpose of this study was to examine a) the relationship between rate of force development (RFD) and vertical jump (VJ) performance during a counter movement jump (CMJ), and b) the reliability of RFD recorded during the CMJ and squat jump (SJ) forms of the VJ. Twenty-three physically active men aged 23 ± 3.9 years participated in the study. Subjects completed 3 unloaded CMJ and 3 unloaded SJ in random order on a force plate. The RFD was measured during CMJ and SJ movements with vertical jump displacement (VJD) measured simultaneously during the CMJ only. Subjects incorporated arm swing to their CMJ technique to reach up as high as possible, and VJD was measured. All SJ were executed with both hands on the hips throughout the full range of movement. Peak rate of force development (PRFD), peak force (PF), and time to peak force (TPF) were significantly correlated to VJD during the CMJ ($r = 0.68$, $r = 0.51$, and $r = -0.48$, respectively). The RFD and TPF during the CMJ and SJ were associated with low test-retest reliability (coefficient of variation [CV]: 11.8–7.9%). Peak and average power, PF, and VJD produced high test-retest reliability (CV: 2.8–5.1%) during both the CMJ and SJ movements. Our results indicate that PRFD, a measure of explosive strength, and PF, a measure of maximal strength, are the primary contributors to VJD during the CMJ in physically active men. However, caution must be used when interpreting data using PRFD because of its low retest reliability.

KEY WORDS countermovement jump, squat jump, reliability, force, power

INTRODUCTION

The vertical jump (VJ) is often used as a performance test to assess athletic ability, identify athletes' strengths and weaknesses, and measure the effectiveness of training programs (13,26). The VJ performance is determined by a complex interaction among several factors, including the maximal force developed by the musculature involved, the rate at which force can be developed, and the neuromuscular coordination of the upper- and lower-body segments (16,27). To measure the contribution of these factors, different protocols and devices have been used, including the use of contact mats, position transducers, V-scopes, accelerometers, rotary encoders, yardsticks, and force plates (4,5,14,21,22,29). Common performance measures calculated from these devices are peak power (PP) or average power (AP). Although force plates are considered the "gold standard" in force measurement (24), force plates are expensive and not easily accessible outside the laboratory setting.

The VJ height has been widely used by sports performance professionals as an alternative to direct assessment of maximal force and power (3,13,27). Recently, the validity of predicting PP and AP on the basis of VJ height has been challenged (5,8,33). Furthermore, the use of the term "power" as a mechanical construct to indicate maximal exercise performance is unclear, with strength qualities such as rate of force development (RFD), impulse, and explosive strength being suggested as a better predictor of athletic ability and performance (6). **The RFD is the development of maximal force in minimal time and is typically used as an index of explosive strength** (36).

Although RFD has been shown to be an important performance variable by some investigators (7,11,31,32), others have reported a poor relationship between RFD and the VJ (9,21,30). **Indeed, Marcora et al. (21) found no significant correlation between RFD and VJ performance measured during an isometric contraction in a horizontal squat position. The lack of a significant relationship between RFD and the VJ may be a result of methodologic problems associated with the measurement of RFD.** Confounding factors include separate tests (30) when examining the relationship between RFD and the VJ and the inclusion of both male and female subjects in the assessment of RFD and VJ performance (9). Recently, Kawamori et al. (17) reported

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a correlation, albeit a nonsignificant one, between dynamic RFD and VJ performance ($r = 0.65$ – 0.74). The lack of significant correlation between RFD and VJ was most likely caused by low statistical power ($n = 8$). To resolve the problem of low statistical power, the present study used 23 males to determine the relationship between RFD and VJ. Therefore, the aim of the present study was to examine a) the relationship between RFD and VJ performance during a counter movement jump (CMJ), and b) the reliability of RFD data recorded during the CMJ and squat jump (SJ) forms of the VJ. It was hypothesized that there would be a significant correlation between RFD and VJ performance, suggesting that the ability to develop force quickly enhances VJ height. However, the use of RFD as a performance measure may not provide the reliability needed for field-based athlete testing.

METHODS

Experimental Approach to the Problem

Subjects attended the Sport Science Laboratory on 2 separate occasions to participate in a familiarization session and a testing session. The test session consisted of a warm-up that included a series of cycle ergometry and dynamic range of movement activities before subjects randomly completed 3 CMJs and 3 SJs on a force plate. During the CMJ, subjects used the stretch shortening cycle and incorporated arm swing into the movement to achieve a maximal jump and reach to record VJ height. During the SJ, subjects started the VJ from a stationary semi squat position with both hands held on the hips throughout the full range of movement. Independent t -tests and correlation coefficients were used to analyze the force-time variables from the force plate during the CMJ and SJ movements. Force-time variables analyzed during the CMJ and SJ included peak force (PF), time to peak force (TPF), peak rate of force development (PRFD), average rate of force development (ARFD), PP, and AP. Vertical jump displacement (VJD) was measured during the CMJ only, determined as the difference between standing reach and jump reach height.

Subjects

Twenty-three physically active men volunteered to participate in this study. None of the men who participated in the present study were experienced in explosive exercise, none were resistance trained, and none of the subjects were participating in regular resistance training before data collection. The subjects were considered to be physically active on the basis that they were actively involved in recreational sports such as Australian Rules Football, soccer, and rugby, in which jumping activities are typical of match play. Subject descriptive data are listed in Table 1. Before participating in the present study, all subjects completed the Physical Activity Readiness Questionnaire and medical history questionnaires and gave written consent in accordance with the guidelines set forth by Bond University

TABLE 1. Descriptive characteristics of subjects ($n = 23$).

Variable	Mean	SD
Age (yr)	23.0	± 3.9
Height (cm)	179.5	± 9.9
Body mass (kg)	81.0	± 10.5

Human Research Ethics Committee. All subjects who participated in the present study attended a familiarization session 3 days before testing. During that familiarization session, subjects received instruction in relation to the performance of the CMJ and SJ and completed the same protocol used in testing. Subjects were asked to refrain from strenuous exercise for 48 hours before the test session.

Procedures

Vertical Jump. Before performing the unloaded CMJ and SJ tests, subjects completed a warm-up consisting of 10 minutes of self-paced cycle ergometry followed by 5 minutes of prescribed dynamic stretching. Once positioned on the force plate, subjects performed 1 submaximal practice jump for

TABLE 2. Performance characteristics during the countermovement jump (CMJ) and squat jump (SJ) ($n = 23$).

Variable	CMJ	SJ
Vertical jump displacement (m)		
Mean	0.55	—
SD	± 0.1	—
Peak force (N)		
Mean	1,972	1,825
SD	± 358	± 273
TPF (ms)		
Mean	239	197
SD	± 65	± 71
PRFD ($N \cdot s^{-1}$)		
Mean	10,728	9,718
SD	$\pm 4,014$	$\pm 3,380$
ARFD ($N \cdot s^{-1}$)		
Mean	4,790	4,375
SD	$\pm 2,155$	$\pm 1,809$
Peak power (W)		
Mean	4,774*	4,056
SD	± 722	± 675
Average power (W)		
Mean	2,359*	1,990
SD	± 456	± 401

TPF = time to peak force; PRFD = peak rate of force development; ARFD = average rate of force development.

*Significant at $p < 0.01$.

TABLE 3. Coefficients of variation (CV), intraclass correlation coefficients (ICC), and associated 95% confidence limits for the force-time variables during countermovement jump ($n = 23$).

Variable	CV%	95% Confidence limits		ICC	95% Confidence limits	
		Lower	Upper		Lower	Upper
Vertical jump displacement (m)	3.3	2.6	4.7	0.98	0.92	0.99
Peak force (N)	3.5	2.8	4.6	0.97	0.91	0.99
TPF (ms)	11.8	9.4	15.8	0.90	0.74	0.97
PRFD ($N \cdot s^{-1}$)	16.3	13.7	20.4	0.89	0.77	0.95
ARFD ($N \cdot s^{-1}$)	17.9	14.2	24.2	0.89	0.72	0.96
Peak power (W)	2.8	2.1	4.0	0.97	0.97	0.99
Average power (W)	3.7	2.8	5.3	0.97	0.93	0.99

TPF = time to peak force; PRFD = peak rate of force development; ARFD = average rate of force development.

both the CMJ and SJ. Each subject then performed 6 VJs (3 CMJs and 3 SJs) beginning with either the CMJ or SJ, then alternating between jumps with 3 minutes of rest between each jump. Displacement for each CMJ was measured with a Vertec (SWIFT Performance Equipment, Lismore, Australia) and has been described previously (9,19). To establish standing reach height, each subject stood side-on to the Vertec jumping device while keeping the heels on the floor and reached upward as high as possible to displace the 0 reference vane.

Each subject began the CMJ in the standing position, dropped into the squat position, and then immediately jumped vertically, incorporating arm swing to jump as high as possible and displacing the vane at the maximum height of the jump. The depth of knee flexion and the amount of arm movement used during each CMJ was individually determined by each subject. Take-off from 2 feet was strictly monitored with no preliminary steps or shuffling permitted during the eccentric or transition phases of the CMJ

technique. The distance between the standing reach and maximum jump height reached on the Vertec to the nearest centimeter (cm) was taken as the VJD for the CMJ. The SJ technique required the subject to descend to a position of 90° knee flexion, determined using a hand-held goniometer, that positioned the upper thigh parallel with the ground (9). Subjects were instructed to hold this position for 3 seconds, after which time the subject jumped for maximum height without prior countermovement. All SJ were executed with both hands on the hips throughout the full range of take off, flight, and landing movements. The best result from each of the CMJ and SJ protocols was used for analysis.

Both the CMJ and the SJ were performed on a force plate (ONSPOT 2000-1), which sampled at a rate of 1,000 Hz, and the analogue signal was converted to a digital signal using a PowerLab 30 series data acquisition system (ADInstruments, Sydney, Australia). The vertical force-time data were filtered using a fourth-order Butterworth low-pass filter with a cutoff frequency of 17 Hz (35).

TABLE 4. Coefficients of variation (CV), intraclass correlation coefficients (ICC), and associated 95% confidence limits for force-time variables during squat jump ($n = 23$).

Variable	CV%	95% Confidence limits		ICC	95% Confidence limits	
		Lower	Upper		Lower	Upper
Peak force (N)	2.8	2.3	3.6	0.97	0.92	0.99
TPF (ms)	13.2	10.6	17.5	0.88	0.70	0.95
PRFD ($N \cdot s^{-1}$)	14.8	11.9	19.5	0.89	0.75	0.96
ARFD ($N \cdot s^{-1}$)	16.9	13.4	22.8	0.89	0.73	0.96
Peak power (W)	3.4	2.6	4.8	0.96	0.91	0.99
Average power (W)	5.1	3.9	7.4	0.94	0.87	0.98

TPF = time to peak force; PRFD = peak rate of force development; ARFD = average rate of force development.

TABLE 5. Intercorrelation of force-time variables during countermovement jump.

		CMJ						
		VJD	PF	TPF	PRFD	ARFD	PP	AP
CMJ	VJD	1.00						
	PF	0.51*	1.00					
	TPF	-0.48*	-0.12	1.00				
	PRFD	0.68†	0.64†	-0.46†	1.00			
	ARFD	0.49*	0.63†	-0.44	0.80†	1.00		
	PP	0.73†	0.83†	-0.11	0.60*	0.48*	1.00	
	AP	0.68†	0.86†	-0.22	0.72†	0.73†	0.90†	1.00

CMJ = countermovement jump; VJD = vertical jump displacement; PF = peak force; TPF = time to peak force; PRFD = peak rate of force development; ARFD = average rate of force development; PP = peak power; AP = average power.

*Significant at $p < 0.05$.

†Significant at $p < 0.01$.

Calculation of Force Variables. The force-time data examined during the CMJs and SJs included PF, PRFD, ARFD, TPF, PP, and AP. A jump was deemed to have started when the vertical force exceeded 10 N greater than the mass of the subject. The PF was calculated as the maximum force achieved over the force-time curve during the jump. The PRFD was calculated from the maximum force that occurred over the first derivative of the force-time curve. The ARFD was calculated as the PF divided by the time taken to achieve the PF. The TPF was taken as the time from the start of the jump until PF was reached. The vertical velocity that was calculated from the integration of the force-time trace was used in the calculation of PP and AP. The vertical force was multiplied by the velocity throughout the propulsive phase of the jump, yielding power. The maximum value was taken as PP with

total work during the propulsion phase divided by the time of the propulsion phase to provide a measure of AP.

Statistical Analyses

The FTVs analyzed during the CMJ and SJ included VJD, PF, TPF, PRFD, ARFD, PP, and AP. The data were represented as mean and *SD*. Independent *t*-tests were used to analyze the FTVs between the CMJ and SJ, with the level of significance set at $p \leq 0.05$. Any significant differences were identified using Tukey's honestly significant difference (HSD) test. Pearson's product moment correlation coefficients were used to determine the relationship between force-time variables and VJ performance. Within-subject variation and reliability for VJ displacement and force-time variables was determined by calculating the coefficient of variation (CV), confidence

limits (95%), and intra class correlation coefficients (ICC) as described by Hopkins (15). All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS for Windows, version 11.0; SPSS, Inc., Chicago, IL, USA).

RESULTS

The mean ($\pm SD$) values for the CMJ and SJ are presented in Table 2. The PP ($p = 0.007$) and AP ($p = 0.006$) were significantly higher during the CMJ than during the SJ. There were no other significant differences in performance measures during the CMJ and SJ. Tables 3

TABLE 6. Intercorrelation of force-time variables during squat jump.

		SJ					
		PF	TPF	PRFD	ARFD	PP	AP
SJ	PF	1.00					
	TPF	-0.28	1.00				
	PRFD	0.76†	-0.67†	1.00			
	ARFD	0.68†	-0.72†	0.94†	1.00		
	PP	0.73†	0.04	0.48*	0.36	1.00	
	AP	0.82†	-0.40	0.72†	0.67†	0.85†	1.00

SJ = squat jump; PF = peak force; TPF = time to peak force; PRFD = peak rate of force development; ARFD = average rate of force development; PP = peak power; AP = average power.

*Significant at $p < 0.05$.

†Significant at $p < 0.01$.

and 4 provide the calculated CV and ICC along with the associated 95% confidence limits for each of the force-time variables recorded during the CMJ and SJ, respectively. For both CMJ and SJ movements, PF, PP, and AP all demonstrated high test-retest reliability (CV range: 2.8–5.1%) and high test-retest correlations (ICC range: 0.91–0.99). The VJD demonstrated high test-retest reliability for the CMJ (ICC: 0.98) with low within-individual variation (CV: 3.3%). The TPF, PRFD, and ARFD for the CMJ and SJ demonstrated low test-retest reliability (CV range: 11.8–17.9%) and test-retest correlations (ICC range: 0.72–0.97).

The interrelationship between force-time variables for the CMJ and SJ are presented in Tables 5 and 6. A significant relationship ($r = 0.68$; $p = 0.001$) between VJD and PRFD can be observed for the CMJ. There were significant relationships between VJD, PP, and AP during the CMJ (Table 5). The TPF during the CMJ was significantly related to PRFD and ARFD, respectively (Table 5), for the SJ; however, during the CMJ, TPF was significantly related to PRFD only (Table 6). The PRFD was also significantly related to PF, ARFD, and AP for both VJ methods (Tables 5 and 6). With the exception of TPF, PF showed significant relationships with all force-time variables measured for both the CMJ and SJ (Tables 5 and 6).

DISCUSSION

The results of the present study suggest that maximal unloaded VJD measured by way of the CMJ is primarily determined by PRFD. In the present study, PRFD had a significant correlation ($r = 0.68$; $p = 0.001$) to VJD. The present results are in contrast with the results of others (9,17,21,31), which reported poor relationships between PRFD and VJ performance. Wilson et al. (31) found no significant relationship between RFD and dynamic performance of the CMJ and suggested that only concentric tests were a superior testing method for RFD. Haff et al. (9) also found no correlation between RFD and CMJ or SJ in collegiate male and female track athletes. However, methodologic differences such as the use of isometric tests to measure PRFD and variations in test methodologies to determine VJD in comparative studies (9, 21, 34) may explain the contrasting results with the results of the present study.

The present study measured VJD while simultaneously measuring PRFD during CMJ and SJ movements on a force plate. The rationale for this methodology was to examine the subjects' ability to develop force rapidly during dynamic movement rather than under conditions of isometric force development, in which a relationship between PF development and its application as an indicator of athletic performance has been challenged (31). Several studies (10,17) have not used arm swing in their CMJ or calculated VJD by flight time. Methods using flight time to estimate jump height are prone to errors associated with the potential for take off and the landing positions to be inconsistent (18). Accordingly, flight time was not considered to determine VJD for the SJ in the present study.

Arm swing during the CMJ was implemented during the present study from a functional perspective whereby the predominance of sport-specific movements and skills require the arms to be incorporated into dynamic movement and swung vigorously upward during takeoff to enhance performance. As such, the use of arm movement during the CMJ was considered a more natural movement for subjects to perform. The use of the arms during VJ testing has been reported to increase the skill and coordination requirements of the movement and therefore may lessen the validity of the VJ as a test of lower-body musculature (8,28). Recent evidence suggests that RFD and muscular strength in lower limbs play a greater role in VJ performance than skill, coordination, any motor learning effect, or familiarization procedures (2,30). In addition, the use of the arms during the VJ has been shown to result in an increase in takeoff velocity when compared with a VJ without arm swing (12, 20). In support of these findings, the present study found a significant negative correlation between TPF and VJD ($r = -0.48$; $p = 0.033$) during the CMJ (Table 5), highlighting the importance of developing muscular force rapidly during a VJ to attain maximal VJD. The increase in takeoff velocity may, in part, explain why a significant correlation ($r = 0.68$; $p = 0.001$) was found between PRFD and VJD during the CMJ in the present study and not in those studies (12,20) in which arm swing was not used.

In addition to PRFD, PF was significantly correlated ($r = 0.51$; $p = 0.023$) to VJD during the CMJ, which suggests that the maximum muscular strength of an individual also contributes to VJD. Petersen et al. (25) found that maximum muscular strength (as measured by 1 repetition maximum back squat) was significantly correlated to VJD in college-aged athletes. Variability in TPF within an individual, however, is most likely caused by changes in neural drive, which have a significant influence over force development during the early (0–50 ms) and mid phase (50–200 ms) of increasing muscle force (1). However, because the TPF in the present study occurred within 250 milliseconds for both the CMJ and the SJ (Table 2), this would suggest that PRFD rather than maximal strength plays a more significant role in VJD (36). The TPF reported in the present study is similar to some studies (10) but not others (17), which reported longer times (>300 ms) to reach PF.

The training status of an individual largely determines the ability to reach PF quickly, with explosively trained athletes achieving the fastest TPF (36). The subjects who participated in this study were all physically active but were untrained in explosive exercise, and, therefore, PRFD exceeded expectations. The results of the present study found poor reliability for PRFD during the CMJ and SJ (Tables 3 and 4). The poor reliability of PRFD data during the VJ in the present study is consistent with the work of Moir et al. (23) and is in contrast with previous studies that have used PRFD during the VJ to assess performance (10,17).

Inconsistency with previous findings (10) may be caused by the inclusion of subjects experienced in training with dynamic explosive exercise and the analysis of a single CMJ and SJ trial after 2 CMJ and VJ warm-up trials. The present study included physically active males with no experience in Olympic style lifts or plyometric exercise. The present study used the best of 3 trials of both the CMJ and SJ for analysis and incorporated arm movement into the CMJ, in contrast with the work of Haff et al. (10), who executed all CMJ and SJ trials with hands on hips. Ambiguity also exists with regard to the sequencing of testing and the inclusion of additional test regimes in previous findings (10) that have used PRFD to assess VJ performance. Kawamori et al. (17) included subjects who were experienced weightlifters and who were well accustomed to explosive exercise and conducted all CMJ and SJ trials with hands on the hip in contrast with the methodology of the present study. VJD, PF, PP, and AP from the CMJ and SJ (Tables 3 and 4) in the present study did show a high degree of reliability and are consistent with previous findings (23).

In summary, the present data indicate that there is a significant correlation between VJD and PRFD during the CMJ movement in young, physically active men. The significant correlation between PRFD and VJD suggests that CMJ performance using the jump and reach method is primarily caused by the ability to develop force rapidly (RFD) and to a lesser extent maximal strength (PF). The use of PRFD as a measure of VJ performance when testing males untrained in dynamic explosive exercise techniques, however, requires caution because of the poor reliability of PRFD and ARFD data.

PRACTICAL APPLICATIONS

The primary aim of the present study was to examine the relationship between RFD and VJD during the CMJ using a Vertec yardstick measurement device on the basis that access to portable force plates and other portable position transducer equipment is often beyond the reach of sports performance professionals outside of the professional sporting environment. Under these simple field-testing conditions, if a positive relationship between VJD and RFD could be determined, scope would exist for athletes, coaches, and other sports professionals to consider RFD as a primary determinant of performance in addition to leg power from VJD.

Of the force-time variables measured during the CMJ movements in this study, 46.4% of VJD was determined by PRFD and as such was the largest contributor to VJD in physically active men. This outcome indicates that individuals who produce greater PRFD will have greater VJ performance. Accordingly, training methods emphasizing explosive technique that are designed to improve PRFD should lead to improvements in VJ and ultimately improved dynamic sports performance. In addition to PRFD, PF was found to contribute 25.6% to VJD during the CMJ, and therefore maximal strength training designed to develop PF should be included in strength training programs to improve VJ

performance. The inclusion of maximal strength training would therefore be appropriate with respect to exercise prescription for untrained or relatively inexperienced individuals for whom the opportunity to improve maximal strength is greater in comparison with elite athletes, who may already be training at or near their genetically predetermined strength capabilities.

The findings of the present study support the inclusion of both explosive-type training with minimal loading to improve PRFD and traditional heavy strength training to enhance PF in physically active but inexperienced strength training individuals to improve VJ performance. Researchers and sports professionals should, however, use caution with respect to the poor reliability of PRFD data to determine VJ performance in untrained individuals.

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